

संभव BATCH

Under the Guidance of Vipulendra Tripathi Sir

Geography: Latitude, Longitude, IST & Solar System | SSC CGL Hard Level MCQs

Q1. Consider the following statements regarding the Solar System:

1. Jovian planets have thick atmospheres consisting mainly of hydrogen and helium.
2. Terrestrial planets were formed in the close vicinity of the parent star where it was too warm for gases to condense.
3. The asteroid belt is located between Jupiter and Saturn.

Which of the statements given above is/are correct?

[SSC CGL Tier-II 2023]

(A) 1 and 2 only

(B) 2 and 3 only

(C) 1 and 3 only

(D) 1, 2 and 3

Q2. Assertion (A): Venus is the hottest planet in our solar system despite Mercury being much closer to the Sun.

Reason (R): Venus has a dense atmosphere composed primarily of carbon dioxide, which creates a runaway greenhouse effect.

[SSC CGL Tier-I 2022]

(A) Both A and R are true and R is the correct explanation of A.

(B) Both A and R are true but R is NOT the correct explanation of A.

(C) A is true but R is false.

(D) A is false but R is true.

Q3. Which of the following statements regarding the planets of our solar system is NOT correct?

[SSC CHSL Tier-I 2021]

- (A) All planets revolve around the Sun in elliptical orbits.
- (B) Uranus rotates on its axis from east to west, similar to Venus.
- (C) Ganymede, the largest moon in the solar system, orbits the planet Saturn.
- (D) Neptune is the farthest known planet from the Sun in our solar system.

Q4. Match List I (Planetary Feature) with List II (Planet):

List I (Feature)	List II (Planet)
A. Highest average density	1. Jupiter
B. Lowest average density	2. Earth
C. Shortest rotation period	3. Venus
D. Longest rotation period	4. Saturn

Select the correct code:

[SSC CPO 2020]

- (A) A-2, B-4, C-1, D-3
- (B) A-4, B-2, C-3, D-1
- (C) A-2, B-4, C-3, D-1
- (D) A-4, B-2, C-1, D-3

Q5. Consider the following statements about Comets, Meteors, and Asteroids:

- 1. Asteroids are rocky and metallic objects mostly found between Mars and Jupiter.**
- 2. A meteor is the visible streak of light that occurs when a meteoroid enters Earth's atmosphere.**
- 3. A comet's tail always points towards the Sun as it approaches the inner solar system.**

Which of the statements given above are correct?

[SSC CGL Tier-I 2021]

(A) 1 and 2 only

(B) 2 and 3 only

(C) 1 and 3 only

(D) 1, 2 and 3

Q6. Assertion (A): We always see the same side of the Moon from the Earth.

Reason (R): The Moon's period of rotation on its axis is exactly equal to its period of revolution around the Earth (tidal locking).

[SSC MTS 2022]

(A) Both A and R are true and R is the correct explanation of A.

(B) Both A and R are true but R is NOT the correct explanation of A.

(C) A is true but R is false.

(D) A is false but R is true.

Q7. Consider the following statements regarding Dwarf Planets according to the IAU definition:

1. A dwarf planet must orbit the Sun.
2. It must have sufficient mass for its self-gravity to pull it into a nearly round shape.
3. It must have cleared the neighbourhood around its orbit.

Which of the statements given above is/are correct?

[SSC CGL Tier-II 2022]

(A) 1 and 2 only

(B) 2 and 3 only

(C) 1 and 3 only

(D) 1, 2 and 3

Q8. Which of the following statements about the 'Goldilocks Zone' (Habitable Zone) is NOT correct?

[SSC CGL 2020]

(A) It is the region around a star where conditions are just right for liquid water to exist on a planet's surface.

(B) Earth is the only planet in our solar system located completely within this zone.

(C) Planets too close to the star will have their water freeze due to extreme solar radiation.

(D) The distance of the Goldilocks zone varies depending on the size and temperature of the star.

Q9. Match List I (Solar System Region) with List II (Characteristics):

List I	List II
A. Oort Cloud	1. Ring of icy bodies beyond Neptune
B. Kuiper Belt	2. Spherical shell of icy debris at the edge of the solar system
C. Asteroid Belt	3. Region between terrestrial and jovian planets
D. Heliosphere	4. Region of space dominated by solar wind

Select the correct code:

[SSC CPO 2022]

(A) A-2, B-1, C-3, D-4

(B) A-1, B-2, C-4, D-3

(C) A-2, B-1, C-4, D-3

(D) A-1, B-2, C-3, D-4

Q10. Consider the following pairs regarding moons and their parent planets:

1. Phobos and Deimos - Mars

2. Titan - Jupiter

3. Triton - Neptune

4. Io - Saturn

How many of the above pairs are correctly matched?

[SSC CGL Tier-I 2023]

(A) Only one pair

(B) Only two pairs

(C) Only three pairs

(D) All four pairs

Q11. Assertion (A): Solar eclipses do not happen every new moon day.

Reason (R): The Moon's orbit is tilted at about 5 degrees relative to Earth's orbit around the Sun.

[SSC CHSL Tier-I 2022]

(A) Both A and R are true and R is the correct explanation of A.

(B) Both A and R are true but R is NOT the correct explanation of A.

(C) A is true but R is false.

(D) A is false but R is true.

Q12. Regarding Kepler's Laws of Planetary Motion, consider the following statements:

1. The orbit of every planet is an ellipse with the Sun at one of the two foci.

2. A line joining a planet and the Sun sweeps out equal areas during equal intervals of time.

3. The square of the orbital period of a planet is inversely proportional to the cube of the semi-major axis of its orbit.

Which of the statements given above is/are NOT correct?

[SSC CGL Tier-II 2023]

(A) 1 only

(B) 2 only

(C) 3 only

(D) 1 and 3 only

Q13. Which of the following statements about the internal structure of the Sun is NOT correct?

[SSC CGL 2021]

- (A) The core is the hottest part of the Sun where nuclear fusion takes place.
- (B) The photosphere is the visible outer layer of the Sun.
- (C) The corona is much cooler than the photosphere and can only be seen during a lunar eclipse.
- (D) Sunspots are temporary phenomena on the photosphere that appear darker than the surrounding areas.

Q14. Match List I (Spacecraft/Mission) with List II (Target Body):

List I	List II
A. Cassini	1. Jupiter
B. Juno	2. Saturn
C. New Horizons	3. Mars
D. Perseverance	4. Pluto

Select the correct code:

[SSC CGL 2022]

- (A) A-2, B-1, C-4, D-3
- (B) A-1, B-2, C-3, D-4
- (C) A-2, B-1, C-3, D-4
- (D) A-1, B-2, C-4, D-3

Q15. Consider the following statements regarding exoplanets:

- 1. Exoplanets are planets that orbit a star outside our solar system.**
- 2. The transit method detects exoplanets by observing the slight dip in a star's brightness when a planet passes in front of it.**
- 3. Proxima Centauri b is an exoplanet located in the closest star system to our Sun.**

Which of the statements given above are correct?

[SSC CHSL 2023]

(A) 1 and 2 only

(B) 2 and 3 only

(C) 1 and 3 only

(D) 1, 2 and 3

Q16. Assertion (A): The distance between two consecutive latitudes remains exactly the same from the Equator to the Poles.

Reason (R): Latitudes are parallel circles that run from east to west and never intersect each other.

[SSC CGL Tier-I 2022]

(A) Both A and R are true and R is the correct explanation of A.

(B) Both A and R are true but R is NOT the correct explanation of A.

(C) A is true but R is false.

(D) A is false but R is true.

Q17. Consider the following countries:

- 1. Brazil**
- 2. Democratic Republic of Congo**
- 3. Kenya**
- 4. Indonesia**

Through which of the above countries does the Equator pass?

[SSC CPO 2021]

(A) 1, 2 and 3 only

(B) 2, 3 and 4 only

(C) 1, 3 and 4 only

(D) 1, 2, 3 and 4

Q18. Match List I (Important Latitude) with List II (Degrees):

List I	List II
A. Tropic of Cancer	1. 66.5° S
B. Tropic of Capricorn	2. 23.5° N
C. Arctic Circle	3. 23.5° S
D. Antarctic Circle	4. 66.5° N

Select the correct code:

[SSC MTS 2021]

(A) A-2, B-3, C-4, D-1

(B) A-3, B-2, C-1, D-4

(C) A-2, B-3, C-1, D-4

(D) A-3, B-2, C-4, D-1

Q19. Which of the following statements about Heat Zones of the Earth is NOT correct?

[SSC CGL 2020]

- (A) The Torrid Zone receives the maximum amount of heat as the mid-day sun is exactly overhead at least once a year on all latitudes in this zone.
- (B) The Temperate Zones experience moderate temperatures because the angle of sun rays goes on decreasing towards the poles.
- (C) The Frigid Zone lies strictly between the Arctic Circle and the Tropic of Cancer in the Northern Hemisphere.
- (D) Places in the Frigid Zone are very cold as the sun does not rise much above the horizon.

Q20. Assertion (A): The length of a degree of longitude decreases progressively from the equator to the poles.

Reason (R): Meridians of longitude converge at the poles and are widest apart at the equator.

[SSC CHSL 2022]

- (A) Both A and R are true and R is the correct explanation of A.
- (B) Both A and R are true but R is NOT the correct explanation of A.
- (C) A is true but R is false.
- (D) A is false but R is true.

Q21. Consider the following statements:

1. The Equator is the only latitude that forms a Great Circle.
2. All longitudes (meridians) form Great Circles when combined with their opposite meridian.
3. A great circle represents the shortest distance between two points on a sphere.

Which of the statements given above are correct?

[SSC CGL Tier-II 2021]

(A) 1 and 2 only

(B) 2 and 3 only

(C) 1 and 3 only

(D) 1, 2 and 3

Q22. Match List I (Phenomenon) with List II (Latitude/Region):

List I	List II
A. Midnight Sun	1. Equator
B. Sun overhead twice a year	2. Arctic and Antarctic Circles
C. Doldrums	3. Torrid Zone
D. Horse Latitudes	4. Sub-tropical high pressure belt

Select the correct code:

[SSC CPO 2023]

(A) A-2, B-3, C-1, D-4

(B) A-3, B-2, C-4, D-1

(C) A-2, B-3, C-4, D-1

(D) A-3, B-2, C-1, D-4

Q23. Assertion (A): Seasonal variations in the length of day and night are zero at the equator and maximum at the poles.

Reason (R): The Earth's axis is inclined at an angle of 66.5 degrees to its orbital plane.

[SSC CGL Tier-I 2022]

(A) Both A and R are true and R is the correct explanation of A.

(B) Both A and R are true but R is NOT the correct explanation of A.

(C) A is true but R is false.

(D) A is false but R is true.

Q24. Consider the following statements regarding the Tropic of Cancer:

1. It passes through exactly 8 states in India.
2. The Mahi River crosses the Tropic of Cancer twice.
3. It represents the southernmost latitude where the sun can be directly overhead.

Which of the statements given above is/are NOT correct?

[SSC CGL 2022]

(A) 1 only

(B) 2 only

(C) 3 only

(D) 1 and 3 only

Q25. Which of the following is NOT a characteristic of the regions located in the mid-latitudes (30° to 60°)?

[SSC CHSL Tier-I 2021]

(A) They experience distinct seasonal changes.

(B) They receive direct, vertical rays of the sun at least once a year.

(C) They are dominated by westerly winds.

(D) They generally experience varied weather patterns due to the mixing of warm and cold air masses.

Q26. Assertion (A): Latitudes are also known as parallels.

Reason (R): The linear distance between two consecutive latitudes slightly increases towards the poles due to the oblateness of the Earth.

[SSC MTS 2023]

(A) Both A and R are true and R is the correct explanation of A.

(B) Both A and R are true but R is NOT the correct explanation of A.

(C) A is true but R is false.

(D) A is false but R is true.

Q27. Consider the following pairs (Latitude vs Country it passes through):

1. Equator - Colombia
2. Tropic of Cancer - Mexico
3. Tropic of Capricorn - Argentina
4. Arctic Circle - Norway

How many of the above pairs are correctly matched?

[SSC CGL Tier-II 2023]

(A) Only one pair

(B) Only two pairs

(C) Only three pairs

(D) All four pairs

Q28. Consider the following statements regarding the International Date Line (IDL):

- 1. The IDL is a straight line coinciding completely with the 180° meridian.**
- 2. A traveler crossing the IDL from east to west gains a day.**
- 3. The IDL deviates around island groups like Kiribati to keep them in the same time zone.**

Which of the statements given above is/are correct?

[SSC CGL 2021]

(A) 1 and 2 only

(B) 3 only

(C) 2 and 3 only

(D) 1, 2 and 3

Q29. Assertion (A): If it is 12:00 Noon at Greenwich (0°), the local time at 30° East longitude will be 2:00 PM.

Reason (R): The Earth rotates from west to east, so places east of Greenwich are ahead in time at a rate of 4 minutes per degree of longitude.

[SSC CPO 2022]

(A) Both A and R are true and R is the correct explanation of A.

(B) Both A and R are true but R is NOT the correct explanation of A.

(C) A is true but R is false.

(D) A is false but R is true.

Q30. Which of the following statements about Longitudes is NOT correct?

[SSC CGL Tier I 2022]

- (A) Longitudes are semi-circles running from pole to pole.
- (B) The Prime Meridian and the 180° longitude together form a full circle that divides the Earth into the Eastern and Western Hemispheres.
- (C) The distance between any two longitudes is roughly 111 km uniformly across the globe.
- (D) There are 360 degrees of longitude in total.

Q31. Match List I (Meridian) with List II (Significance):

List I	List II
A. 0° Longitude	1. International Date Line
B. 180° Longitude	2. Indian Standard Time
C. 82.5° E Longitude	3. Prime Meridian (Greenwich)
D. 11.5° E Longitude	4. Central European Time

Select the correct code:

[SSC CHSL 2022]

- (A) A-3, B-1, C-2, D-4
- (B) A-1, B-3, C-2, D-4
- (C) A-3, B-1, C-4, D-2
- (D) A-1, B-3, C-4, D-2

Q32. Consider the following statements:

1. Russia has the highest number of time zones in the world.
2. France, including its overseas territories, spans the maximum number of time zones for any country.
3. China officially follows a single time zone despite its large longitudinal extent.

Which of the statements given above are correct?

[SSC CGL Tier-II 2022]

(A) 1 and 2 only

(B) 2 and 3 only

(C) 1 and 3 only

(D) 1, 2 and 3

Q33. Assertion (A): The local time of places on the same longitude is the same everywhere on Earth.

Reason (R): Places on the same longitude experience midday (sun at the highest point) at exactly the same time.

[SSC CGL 2020]

(A) Both A and R are true and R is the correct explanation of A.

(B) Both A and R are true but R is NOT the correct explanation of A.

(C) A is true but R is false.

(D) A is false but R is true.

Q34. Which of the following countries does the Prime Meridian (0° Longitude) NOT pass through?

[SSC CPO 2021]

(A) United Kingdom

(B) Spain

(C) Mali

(D) Nigeria

Q35. If a cricket match starts in London (GMT) at 10:00 AM on a Monday, at what local time will a viewer in Sydney (150° E) watch the live start?

[SSC CGL Tier-I 2021]

(A) 10:00 PM on Monday

(B) 8:00 PM on Monday

(C) 10:00 AM on Tuesday

(D) 8:00 AM on Monday

Q36. Consider the following statements regarding the crossing of the International Date Line (IDL):

- 1. Moving from Japan to Hawaii (West to East across the IDL), one repeats a calendar day.**
- 2. Moving from California to Australia (East to West across the IDL), one loses a calendar day.**
- 3. The IDL passes exactly through the Bering Strait without touching any landmass.**

Which of the statements given above is/are correct?

[SSC CGL Tier-II 2023]

(A) 1 and 2 only

(B) 2 and 3 only

(C) 1 and 3 only

(D) 1, 2 and 3

Q37. Assertion (A): Standard time is necessary for countries with a large latitudinal extent.

Reason (R): Large latitudinal extent causes massive differences in local times across the country, confusing train and flight schedules.

[SSC CHSL 2020]

(A) Both A and R are true and R is the correct explanation of A.

(B) Both A and R are true but R is NOT the correct explanation of A.

(C) A is true but R is false.

(D) A is false but R is true.

Q38. Which of the following correctly describes 'Antipodal points' on the globe?

[SSC MTS 2023]

- (A) Two points situated exactly on the same longitude but opposite hemispheres.
- (B) Two points located diametrically opposite to each other on the Earth's surface.
- (C) Two points on the equator separated by exactly 90 degrees.
- (D) Places that experience exactly the same weather conditions year-round.

Q39. Consider the following statements regarding Indian Standard Time (IST):

1. It is calculated on the basis of the 82.5° E longitude.
2. It is 5 hours and 30 minutes ahead of Greenwich Mean Time (GMT).
3. The standard meridian passes through Mirzapur in Uttar Pradesh.

Which of the statements given above are correct?

[SSC CGL Tier-I 2022]

- (A) 1 and 2 only
- (B) 2 and 3 only
- (C) 1 and 3 only
- (D) 1, 2 and 3

Q40. Assertion (A): There is a time difference of approximately two hours between the easternmost part of Arunachal Pradesh and the westernmost part of Gujarat.

Reason (R): India has a vast longitudinal extent of about 30 degrees.

[SSC CGL 2023]

- (A) Both A and R are true and R is the correct explanation of A.
- (B) Both A and R are true but R is NOT the correct explanation of A.
- (C) A is true but R is false.
- (D) A is false but R is true.

Q41. Through how many Indian states does the Indian Standard Meridian (82.5° E) pass?

[SSC CPO 2021]

(A) 4 States

(B) 5 States

(C) 6 States

(D) 7 States

Q42. Match List I (State) with List II (Does IST pass through it?):

List I (State)	List II (Status)
A. Madhya Pradesh	1. Yes
B. Telangana	2. No
C. Chhattisgarh	3. Yes
D. Jharkhand	4. No

Select the correct logical matching:

[SSC CHSL Tier-I 2021]

(A) A-1, B-2, C-1, D-2

(B) A-2, B-1, C-2, D-1

(C) A-1, B-1, C-1, D-2

(D) A-1, B-2, C-2, D-1

Q43. Which of the following statements about the intersection of the Tropic of Cancer and the Indian Standard Meridian is correct?

[SSC CGL Tier-II 2022]

- (A) They intersect in the state of Madhya Pradesh.
- (B) They intersect in the Koriya district of Chhattisgarh.
- (C) They intersect in the Mirzapur district of Uttar Pradesh.
- (D) They do not intersect anywhere on the Indian landmass.

Q44. Consider the following statements regarding the proposal for two time zones in India:

1. The proposal suggests 'Chai Bagaan Time' for the northeastern states to save daylight.
2. A dual time zone would lead to the exact alignment of biological clocks across the entire country.
3. Historically, India had two time zones (Bombay Time and Calcutta Time) during British rule.

Which of the statements given above is/are correct?

[SSC CGL 2021]

- (A) 1 and 2 only
- (B) 1 and 3 only
- (C) 2 and 3 only
- (D) 1, 2 and 3

Q45. Assertion (A): Sri Lanka follows the same Standard Time as India (IST).

Reason (R): The 82.5° E longitude passes physically through the central highlands of Sri Lanka.

[SSC MTS 2022]

(A) Both A and R are true and R is the correct explanation of A.

(B) Both A and R are true but R is NOT the correct explanation of A.

(C) A is true but R is false.

(D) A is false but R is true.

Q46. Which of the following states is NOT crossed by the Indian Standard Meridian (82.5° E)?

[SSC CGL Tier-I 2020]

(A) Odisha

(B) Andhra Pradesh

(C) Chhattisgarh

(D) Maharashtra

Q47. Consider the following events:

If an international virtual meeting is scheduled at 2:00 PM GMT, at what local time should an executive in New Delhi and another in Tokyo (135° E) log in respectively?

[SSC CPO 2023]

(A) 7:30 PM and 11:00 PM

(B) 7:30 AM and 11:00 AM

(C) 6:30 PM and 10:00 PM

(D) 7:30 PM and 9:00 PM

Q48. Assertion (A): The standard meridian of a country is generally chosen in multiples of 7.5° .

Reason (R): It ensures that the time difference between the standard time of that country and Greenwich Mean Time is expressed in multiples of half-hours.

[SSC CGL Tier-II 2023]

(A) Both A and R are true and R is the correct explanation of A.

(B) Both A and R are true but R is NOT the correct explanation of A.

(C) A is true but R is false.

(D) A is false but R is true.

Q49. Which of the following correctly represents the approximate longitudinal extent of mainland India?

[SSC CHSL 2021]

(A) $68^\circ 7'$ E to $97^\circ 25'$ E

(B) $68^\circ 7'$ E to $95^\circ 25'$ E

(C) $70^\circ 7'$ E to $97^\circ 25'$ E

(D) $68^\circ 7'$ N to $97^\circ 25'$ N

Q50. Consider the following statements:

- 1. If India were to adopt two time zones, the dividing line is proposed to be the 89° E longitude.**
- 2. Andaman and Nicobar Islands currently follow a time zone half an hour ahead of IST.**
- 3. Lakshadweep Islands strictly follow IST despite being geographically west of the mainland's center.**

Which of the statements given above is/are NOT correct?

[SSC CGL Tier-I 2022]

(A) 1 only

(B) 2 only

(C) 1 and 3 only

(D) 2 and 3 only